

CTET 2023 - Paper I (Mathematics)

(Synthetic practice paper for AI testing - not an actual exam leak)

Q1. [Multiplication] What is the value of 6×12 ?

- (A) 48
- (B) 56
- (C) 42
- (D) 54

Answer: (A)

Q2. [LCM and GCD] The LCM of 7 and 14 is:

- (A) 60
- (B) 30
- (C) 45
- (D) 90

Answer: (A)

Q3. [Percentage] If 8% of a number is 16, what is the number?

- (A) 100
- (B) 80
- (C) 120
- (D) 160

Answer: (A)

Q4. [Fractions] Which fraction is larger: $\frac{2}{3}$ or $\frac{3}{4}$?

- (A) $\frac{3}{4}$
- (B) $\frac{2}{3}$
- (C) equal
- (D) cannot determine

Answer: (A)

Q5. [Geometry] A rectangle has length 10 cm and breadth 20 cm. Its perimeter is:

- (A) 24 cm
- (B) 30 cm
- (C) 36 cm
- (D) 48 cm

Answer: (B)

Q6. [Algebra] If $x + 11 = 22$, what is x ?

- (A) 3
- (B) 5
- (C) 7
- (D) 9

Answer: (B)

Q7. [Time and Distance] A train travels 12 km in 24 hours. Its speed is:

- (A) 60 km/h
- (B) 80 km/h
- (C) 100 km/h
- (D) 120 km/h

Answer: (A)

Q8. [Number System] Which of the following is a prime number?

- (A) 11
- (B) 15
- (C) 21
- (D) 25

Answer: (A)

Q9. [Mensuration] The area of a square with side 14 cm is:

- (A) 25 cm^2
- (B) 36 cm^2
- (C) 49 cm^2
- (D) 64 cm^2

Answer: (A)

Q10. [Ratio and Proportion] The ratio 2:4 is equivalent to:

- (A) 1:2
- (B) 2:3
- (C) 3:4
- (D) 4:5

Answer: (A)

Q11. [Pedagogy of Mathematics] Which approach is BEST suited for introducing fractions to Class 5 students?

- (A) Concrete-Pictorial-Abstract method
- (B) Rote memorization
- (C) Lecture only
- (D) Drill exercises

Answer: (A)

Q12. [Pedagogy] According to NCF 2005, the role of a math teacher is to:

- (A) Facilitate exploration and inquiry
- (B) Deliver only lectures
- (C) Drill formulas
- (D) Test memorization

Answer: (A)

Q13. [Number System] What is the place value of 7 in 5,734?

- (A) 700
- (B) 70
- (C) 7000
- (D) 7

Answer: (A)

Q14. [LCM and GCD] The HCF of 24 and 36 is:

- (A) 12
- (B) 6
- (C) 18
- (D) 24

Answer: (A)

Q15. [Multiplication] 9 multiplied by 11 equals:

- (A) 99
- (B) 98
- (C) 100
- (D) 81

Answer: (A)